Jungwoo Joshua Lee

jungwoojoshualee0424@gmail.com

Education

Chungnam Samsung High School, South Korea — Mar 2024 – Feb 2027

- IB Diploma Candidate (November Exam)
- IB Physics HL, IB Mathematics HL, IB Chemistry HL

Tangjeong Middle School, South Korea — Mar 2021 – Feb 2024

- Completed 9th Grade. South Korean Curriculum

Extracurricular Activities

Solar-Powered RC Car Construction – South Korea — Mar 2026 – Present

Founder & Lead Instructor

- Designed and built modular solar-powered remote-controlled car kits utilizing 3D printing techniques and custom Arduino circuits.
- Taught renewable energy concepts and sustainable engineering principles to local youth through hands-on kit assembly campaigns and instructed using simplified wiring structures.

"Algorable" Computer Science Club – South Korea — Mar 2025 – Present

President & Lead Developer

- Directed student development teams in C/C++ and Python programming projects, including using genetic algorithms for autonomous vehicle simulations.
- Engineered and programmed a custom document converter tool/program to resolve file incompatibility issues across campus workstations to aid in efficiency.
- Produced and broadcast video tutorials online documenting self-taught microcontroller programming and circuit design.

Academic Peer Tutor – South Korea — Mar 2025 – Present

Tutor & Community Leader

- Conducted weekly problem-solving tutoring sessions as the designated peer tutor for IB Physics HL and IB Mathematics HL.
- Coordinated and organized a specialized and skilled team of peer tutors to distribute the workload of students evenly among the volunteers.

3D-Printed Aerodynamics Program – South Korea — Mar 2025 – Feb 2026

Project Lead & Technical Instructor

- Engineered model airplane kits using CAD tools and systematic aerodynamic testing.
- Taught aerodynamics and 3D-printing techniques to elementary students at local libraries, supplemented by published step-by-step video tutorials that were also made available online.
- Produced and filmed video tutorials for beginners and non-engineering students to master 3D-printing.

Self-taught Arduino, Programming, and 3D Printing – South Korea — Jan 2025 – Apr 2026

Independent Learning

- Since IB is not engineering-focused, I was provided with less access to engineering education and tools, so I decided to self-learn through YouTube and Fusion 360, and I wrote an in-depth journal about the education.
- Self-learning led to various projects, such as the solar-powered Arduino RC Car Kit and Aerodynamic Plane.

TRIZ Analysis Journal – South Korea — Sept 2025 – Nov 2025

Independent Researcher

- Executed a 7-week mechanical analysis program evaluating 21 everyday devices using the 40 TRIZ Inventive Principles to identify mechanisms found in inventions in our daily lives.
- Each of the 21 devices was documented in a journal.

Kendo – South Korea — Mar 2024 – Feb 2026

Member

- Practiced once a week for two hours and developed discipline, focus, perseverance, and composure through regular training and competitive sparring.
- During a competitive sparring session, my bamboo sword broke and struck my head; even though I was bleeding, I remained composed and completed the match. This strengthened my understanding of perseverance and the spirit of Kendo.

American Football – South Korea Mar 2026 – July 2026
Linebacker / Wide Receiver
- ❖ Adapted to the unfamiliar demands, rules, and position-specific techniques of American Football and developed a deeper understanding of teamwork and communication through a sport where every player's role directly affects the team's performance.

Asan Ginkgo Road National Marathon – South Korea Mar 2025 – Oct 2025
Participant
- ❖ Trained three times a week over six months to prepare for this marathon while balancing more than 12 hours of daily classes and independent studying.
- ❖ Competed in the Asan Ginkgo Road National Marathon 10K in 1 hour 4 minutes, demonstrating strong perseverance and commitment.

Campus Infrastructure & Woodworking Project – South Korea Aug 2024 – Dec 2024
Lead Fabricator
- ❖ Founded a student-led initiative to revitalize underused common spaces on campus into a collaborative student lounge.
- ❖ Drafted architectural blueprints and manufactured wooden furniture (conference tables, benches, and lamp fixtures) using carpentry, power tools, and soldering.

Class Mathematics Event– South Korea Aug 2024 – Dec 2024
Member
- ❖ Organized a school STEM festival including logic competitions, cryptography design, and binary puzzles.
- ❖ Mapped out guest flow and set up boundaries to ensure organized lines and traffic flow.

Community Service

Asan City Volunteer Center – South Korea Jul 2026
Volunteer
- ❖ Prepared and operated a volunteer-based education program called "Build a Whizzing RC Car to Protect the Earth." Where patrons were taught through hands-on methods about RC cars, renewable energy, and engineering.

Eco-Friendly Fertilizer Project – South Korea Mar 2025 – Feb 2026
Co-founder
- ❖ Developed organic fertilizer formulations using recycled soil and natural minerals to bolster plant growth without compromising sustainability.
- ❖ Organized community workshops to distribute potted plants and eco-friendly fertilizers to local elementary school children to foster early environmental stewardship.

Multicultural Youth Mentorship – South Korea Nov 2025 – Dec 2025
Volunteer Educator
- ❖ Tutored local children from multicultural backgrounds in STEM concepts, Korean language, and English language.
- ❖ Designed and crafted custom hands-on learning kits to drive engagement and build academic confidence from a young age.

Asan City Volunteer Center – South Korea Jul 2025
Environmental Education Volunteer
- ❖ Prepared and operated the youth volunteer program "Coffee, Please Become Soil!" where youth were able to connect coffee waste recycling with environmental education.

Campus Notification Boards – South Korea Apr 2025 – Aug 2025
Lead Designer
- ❖ Designed and illustrated graphics for the unused notification board on campus for the student body to update monthly events, and promoted a multicultural section to update the student body on global cultures and current events happening overseas.

Selected Academic Research

Mathematics Internal Assessment: Dynamic Behavior Modeling 12th Grade

- ❖ Modeled a quarter-car suspension system using 2nd-order differential equations, state-space systems, and phase portraits to analyze underdamped, overdamped, and critically damped behavior.

Physics Extended Essay: Kinematics & Oscillation Modeling 11th – 12th Grade

- ❖ Designed an automated test bench to model the quantitative relationship between driving points of oscillation and instantaneous centers of rotation.
- ❖ Applied Python-based Fourier transform analysis to filter nonlinear noise in physical rod models.

Physics Internal Assessment: Fluid Dynamics Investigation 11th – 12th Grade

- ❖ Constructed a custom acrylic liquid tank system to investigate volumetric flow rates against pipe radius and length using Darcy-Weisbach and Reynolds equations.

Chemistry Internal Assessment: Thermodynamic Equilibrium Dynamics 11th – 12th Grade

- ❖ Experimented with temperature-dependent equilibrium shifts using UV-Vis spectrophotometry, analyzing results by linearizing Van't Hoff logarithmic equations, and resolving an unexpected iron hydrolysis side reaction.

Awards and Achievements

Academic Excellence Award (IB STEM & Core) – Chungnam Samsung High School 12th Grade

- ❖ IB Physics HL, IB Economics SL, IB English B SL, IB Math AI HL

Academic Performance Excellence Award – Chungnam Samsung High School 11th Grade

- ❖ IB Economics SL, IB TOK, IB Chemistry HL

Academic Performance Excellence Award– Chungnam Samsung High School 11th Grade

- ❖ IB TOK

Subject Excellence Award– Chungnam Samsung High School 11th Grade

- ❖ IB Physics HL, IB TOK

1st Place, English Speaking Contest – Chungnam Samsung High School 10th Grade

2nd Place, Invention Contest – Chungnam Samsung High School 10th Grade

2nd Place (Silver), Mathematics Research Contest – Chungnam Samsung High School 10th Grade

2nd Place, Mathematics Research & Inquiry Contest – Tangjeong Middle School 9th Grade

3rd Place, Science Inquiry Contest (Newspaper) – Tangjeong Middle School 9th Grade

2nd Place, English Storytelling Competition – Tangjeong Middle School 9th Grade

Additional Skills

Technical: CAD, Blender, 3D Printing, C/C++, Python (Data Analysis, NumPy/Matplotlib), Arduino, Prototyping (Soldering, Woodworking, and Acrylic Fabrication)

Languages: English (Native), Korean (Native), Japanese (Intermediate)